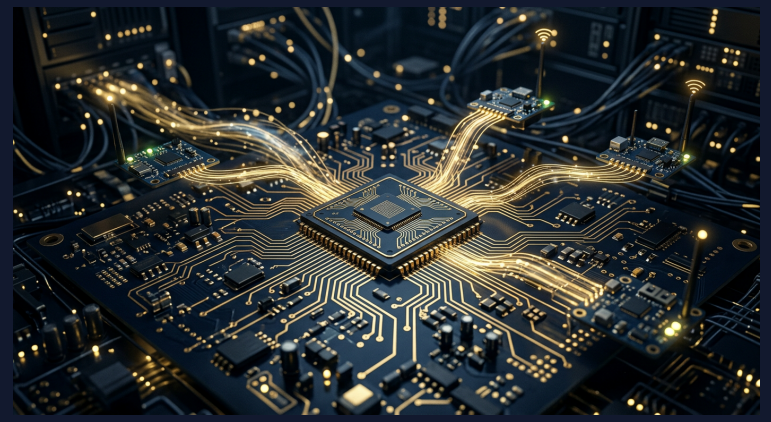
AI.201

Internet of Things

COURSE OVERVIEW

Networked physical devices: sensors, connectivity, cloud integration, and edge computing for smart systems.



ILLUSTRATIVE HERO IMAGERY

INTERNET OF THINGS

1

SESSION

60 MINUTES

ORIENTATION: THE CONNECTED WORLD

KEY TOPICS

- What is IoT
- Sensors, actuators, and things
- Connectivity options
- IoT architecture

TEACHING FOCUS

See the world as a network of sensing devices.

KEY TAKEAWAY IoT is the internet extended to physical things.

2

SESSION

 60 MINUTES

PROTOCOLS & CONNECTIVITY

KEY TOPICS

- MQTT, CoAP, HTTP
- Wi-Fi, Bluetooth, Zigbee, LoRaWAN
- IPv6 and 6LoWPAN
- Gateway and edge devices

TEACHING FOCUS

Choose the right network for each device.

▲ **KEY TAKEAWAY** Connectivity is chosen by range, power, and bandwidth

INTERNET OF THINGS

3

SESSION

 60 MINUTES

CLOUD & EDGE PLATFORMS

KEY TOPICS

- Cloud IoT platforms
- Edge computing and fog
- Data ingestion and storage
- Dashboards and visualization

TEACHING FOCUS

Process data where it makes sense.

▲ **KEY TAKEAWAY** The cloud stores; the edge acts fast.

INTERNET OF THINGS

4

SESSION

60 MINUTES

SECURITY & PRIVACY

KEY TOPICS

- Device authentication
- Encryption in transit and at rest
- Firmware updates
- Privacy by design

TEACHING FOCUS

Secure devices from chip to cloud.

KEY TAKEAWAY A chain is only as strong as its weakest node.

INTERNET OF THINGS

5

SESSION

60 MINUTES

IOT IN ROBOTICS

KEY TOPICS

- Connected robots and fleets
- Telemetry and remote monitoring
- Over-the-air updates
- Case studies

TEACHING FOCUS

Network robots for fleet intelligence.

KEY TAKEAWAY Connected robots learn from each other.

INTERNET OF THINGS

6

SESSION

60 MINUTES

IOT PROJECT & REVIEW

KEY TOPICS

- Mini IoT prototype
- Course recap and assessment
- Toward CPS (AI.302)
- Design review

TEACHING FOCUS

Build a small connected sensing system.

KEY TAKEAWAY Prototyping turns concepts into working things.